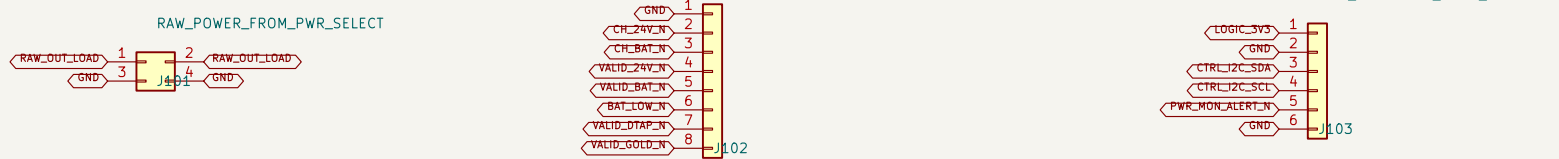


A. PWR-SELECT INTERFACE

D. DETAILED CAPTURE SUITE – REV A1

E. CONNECTED CM5-CARRIER HIERARCHY



Mates with PWR-SELECT (301) two contacts per priority, realized 10 A drawing.

Mates with PWR-SELECT (401) add 3.3 V pull-ups on the carrier.

Mates with PWR-SELECT (402) carrier supplies 3.3 V and even I2C pull-ups.

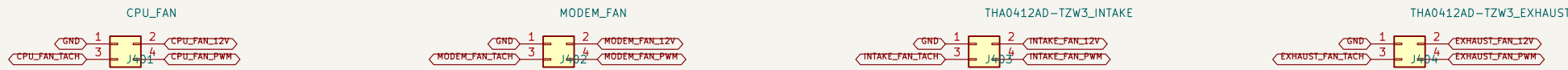
B. BUFFERED DIFFERENTIAL TDM + AUDIO CONTROL



Mates with AUDIO-B08 (351) using Nixie #7832-6433 headers; harness assembly remains vibration-qualified.

Carrier drives HCU/RCU/FSMC/SEC data; AUDIO-B08 drives ADC data back.

C. FOUR INDEPENDENT PWM / TACH FANS



CPU_FAN

M02EM_FAN

TH04L1240-TZW3_INTAKE

TH04L1240-TZW3_EXHAUST

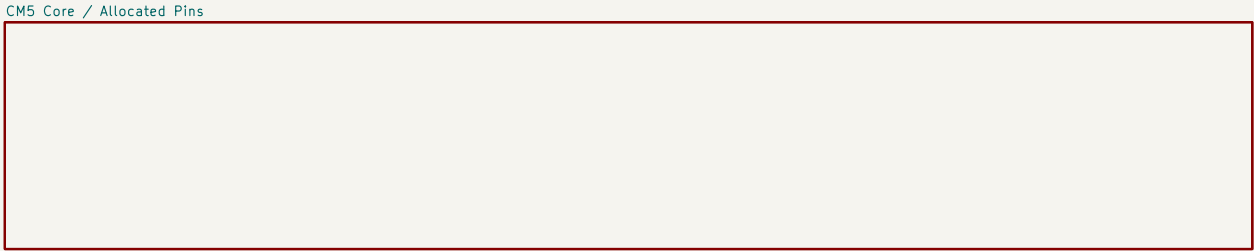
IN33 is physically keyed/tabbed INTAKE; IN46 is keyed/tabbed EXHAUST.

- 01 CM5-Carrier-Allocated exact 301-contact CM5 symbol, 76 named contacts (74 connected, 2 assigned NC)
- 02 CM5-Carrier-Allocated: VCC_SYSN, reset, recovery, power-on and debug UART
- 03 Network-PCie: 87C92G0809F, 3x LAN7430 and native WMI front end
- 04 Network-PCie: for Marlin Magjack and Nixie 0879010102 Mini PCie Nixie Wi-Fi
- 05 WMAN-SIM: TE 219W20-3 8-key, USB3/USB2, controls and dual-SIM mux
- 06 Display-Harness HDMI, USB Touch and 446464E 1.2 V / 2.0 A branch
- 07 Audio-Control: I2S0 TDM, I2S1 E8856 headset and CTA amplifier path
- 08 Thermal-ID: I2C translation, GPD expansion, sensors and four fan channels
- 09 Power-Regulators-AI: protected raw hold-up, all main rails, sequencing and test points
- 10 Audio-B08: all XLR channels bonded to chassis with one controlled AGND bond
- 11 Reserve: all three board tests have zero EDC violations; child control is allowed

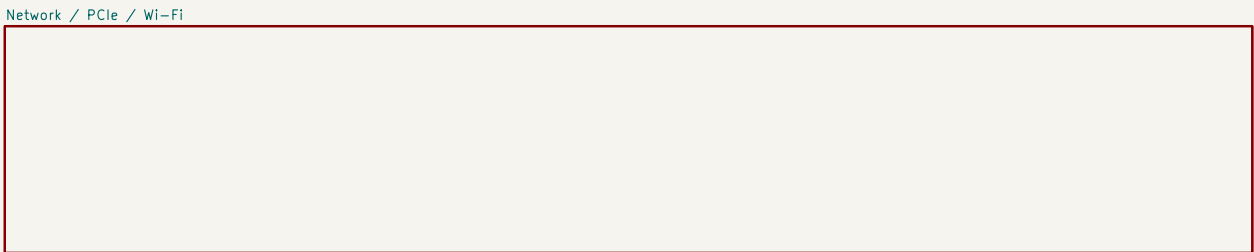
DC/DC compensation and magnetic remote a separate power-design calculation milestone.

High-speed routing starts only after startup and connector Z data are released.

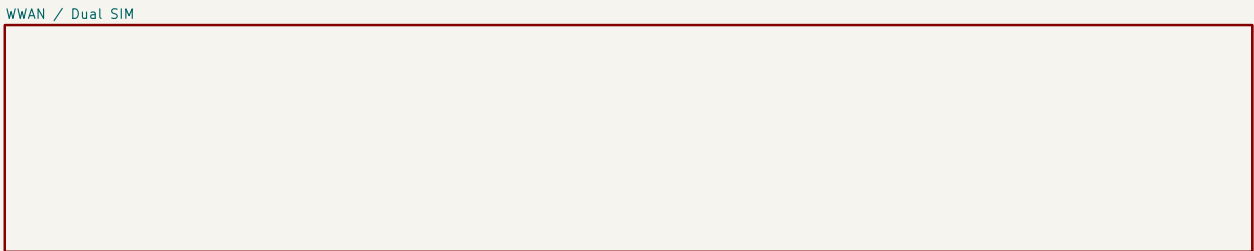
The two enclosure fans require an underside baffle to prevent direct intake/exhaust short flow.



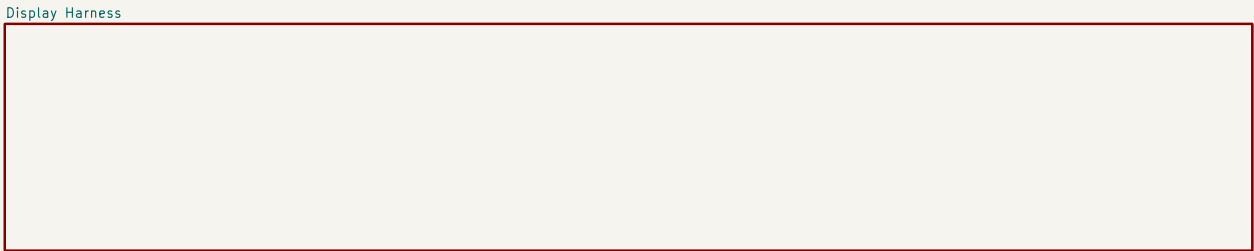
File: CMS-Carrier-Allocated.kicad_sch



File: Network-PCie.kicad_sch



File: WMAN-SIM.kicad_sch



File: Display-Harness.kicad_sch



File: Audio-Control.kicad_sch



File: Power-Regulators-AI.kicad_sch



File: Thermal-ID.kicad_sch

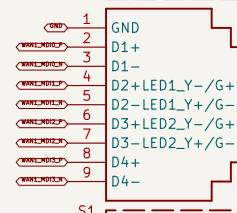
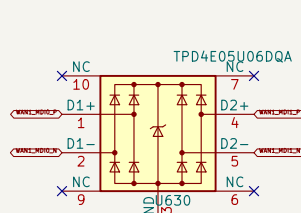
USE/DOE/MS-1046 are Interface-only Application; the detailed-page connections over the board footprint.

[illegible]

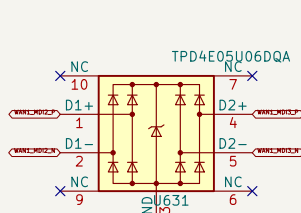
DETAILED CAPTURE BASELINE – verify CM5 voltage limits before release
 U33-A includes the four official grounded CM5 mounting pads; allocated GPIO domain is 3.3V
 76 contacts are owned: 74 connected and 2 assigned no-connect; all other functions are for
 Three DF40C-100DS mates; pad geometry and relative placement come from Radxa IO V2.20
ProComm
 Sheet: /CM5 Core / Allocated Pins/
 File: CM5-Core-Allocated.kicad_sch
Title: Radxa CM5 ProComm Carrier – Exact CM5 Connector Allocation
 Size: A2 Date: 2026-08-13 Rev: A1
 KiCad E.D.A. 10.0.5-rc1 Id: 2/8



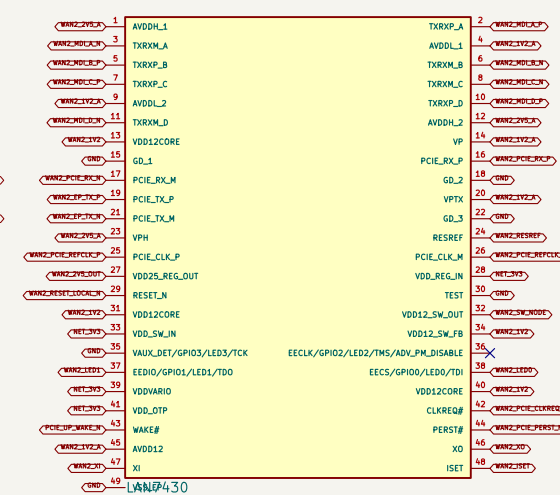
LAN7A30 endpoint pinout, see LAN7A30 pinout table for details. Pin 100 is not used.



74991114412

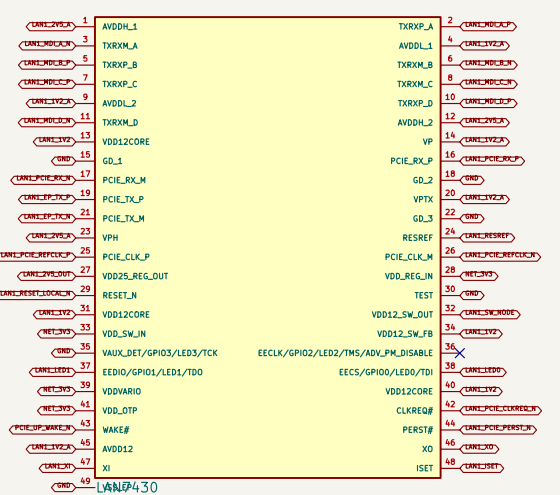


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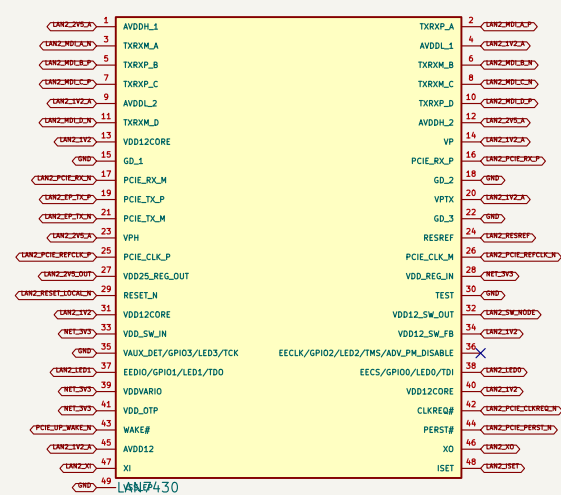
LAN7A30

U611



LAN7A30

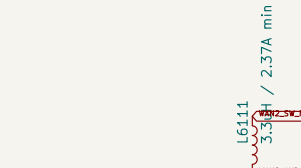
U612



LAN7A30

U613

LAN7A30 endpoint pinout, see LAN7A30 pinout table for details. Pin 100 is not used.



LAN7A30

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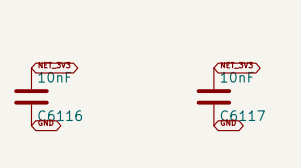
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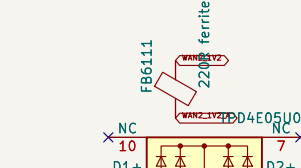
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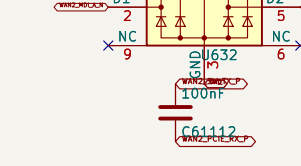
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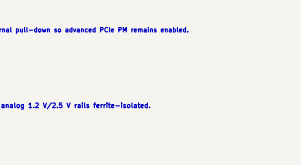
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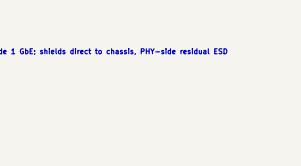
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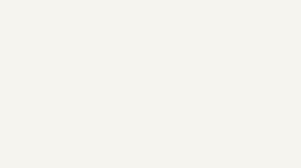
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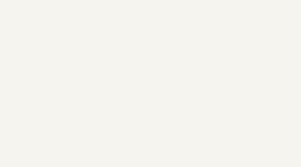
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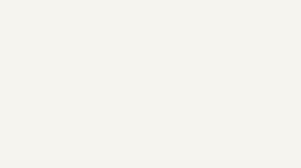
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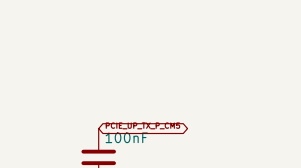
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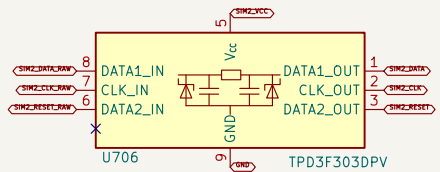
U619

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4. TWO FIELD-ACCESSIBLE NANO-SIM HOLDERS



The crystal design tracks M000M_UIM_PWL, preventing a fixed 3.3 V condition when the SIM rail is 1.8 V.

6.8nF COG

220µF COG

C704

C705

Physical SIM 1 uses FSA2563 channel 2 and physical SIM 2 uses channel 1 to implement that default.



SIM2: TPO3F363 Filters RESET/CLK/DATA and clamps VCC beside the holder; shell bonds to CHASSIS_GND



5. RF BULKHEAD HARNESS CONTRACT



Four off-board (CF 818233349) 25S RF IV-ta-SMA bulkhead pigtails; final cable length remains a routed sample cable

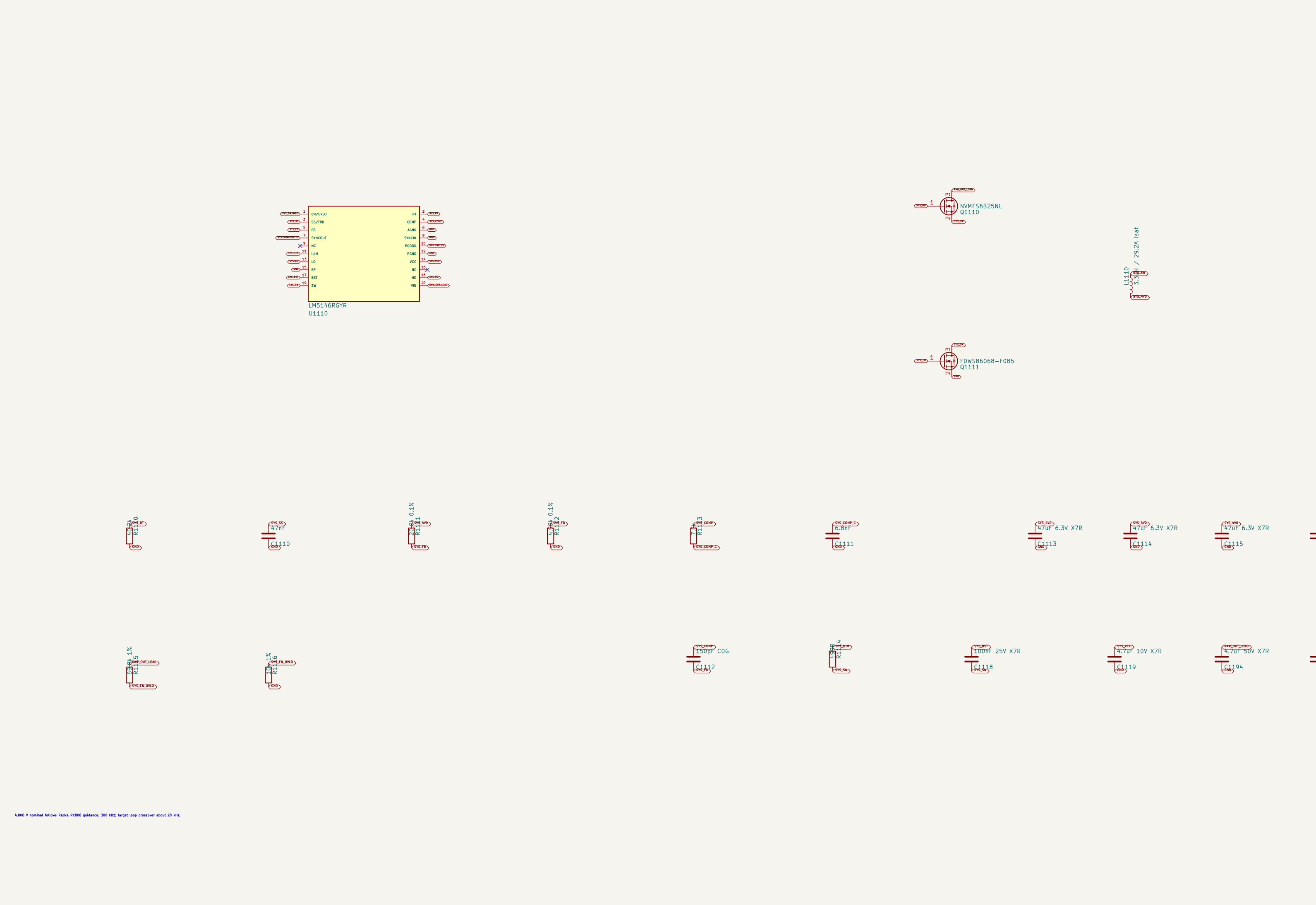
TP7201-TP7208 are copper factory probes, USB 2/3 has no branch or generic test header

ProComm
Sheet: /WWAN / Dual SIM/
File: WWAN-SIM.kicad_sch

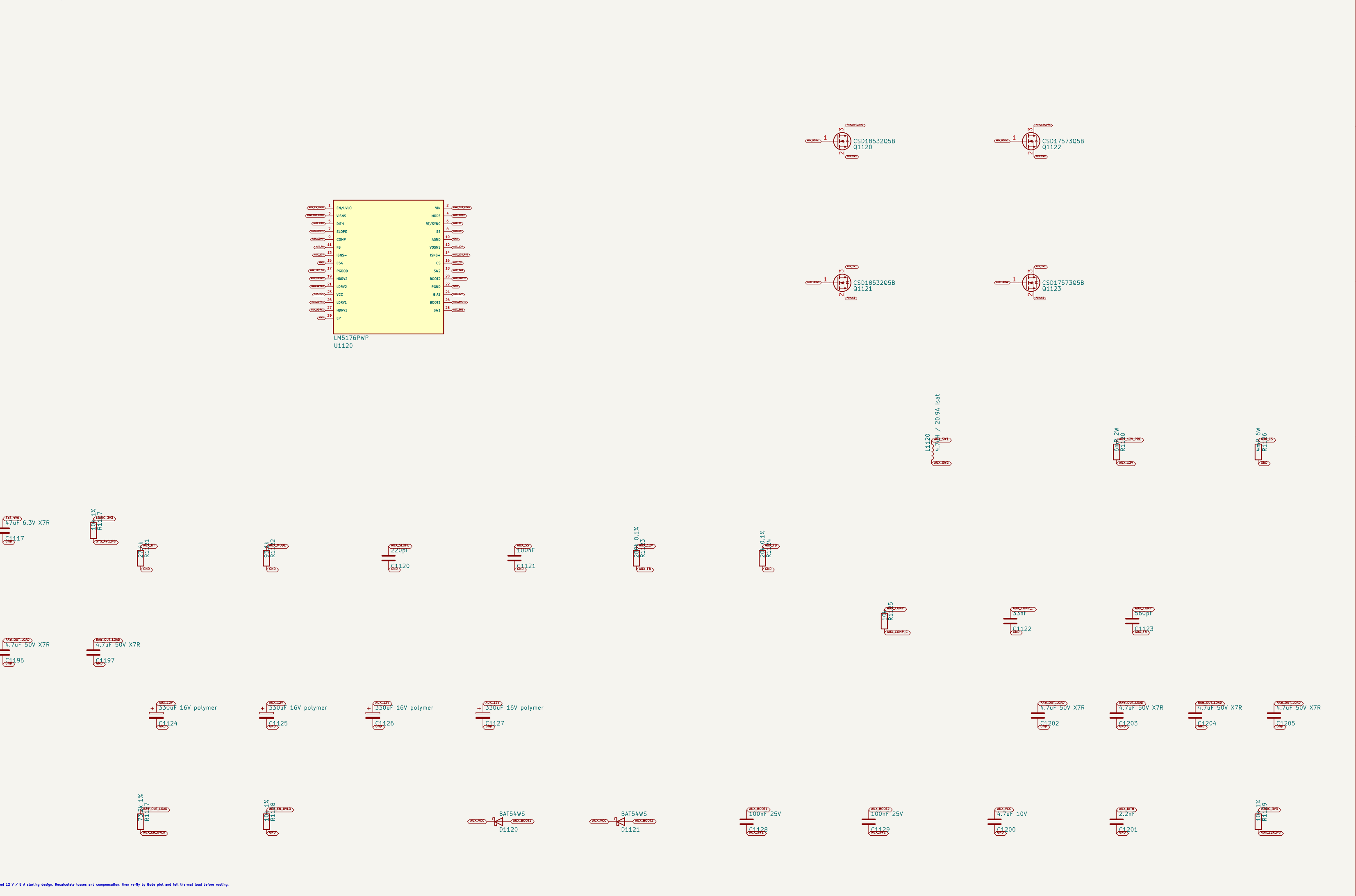
Title: Radxa CM5 ProComm Carrier – WWAN and Dual SIM		
Size: A1	Date: 2026-08-14	Revised:
KiCad E.D.A. 10.0.5-rc1		Id:

1. PROTECTED RAW INPUT FROM PWR-SELECT

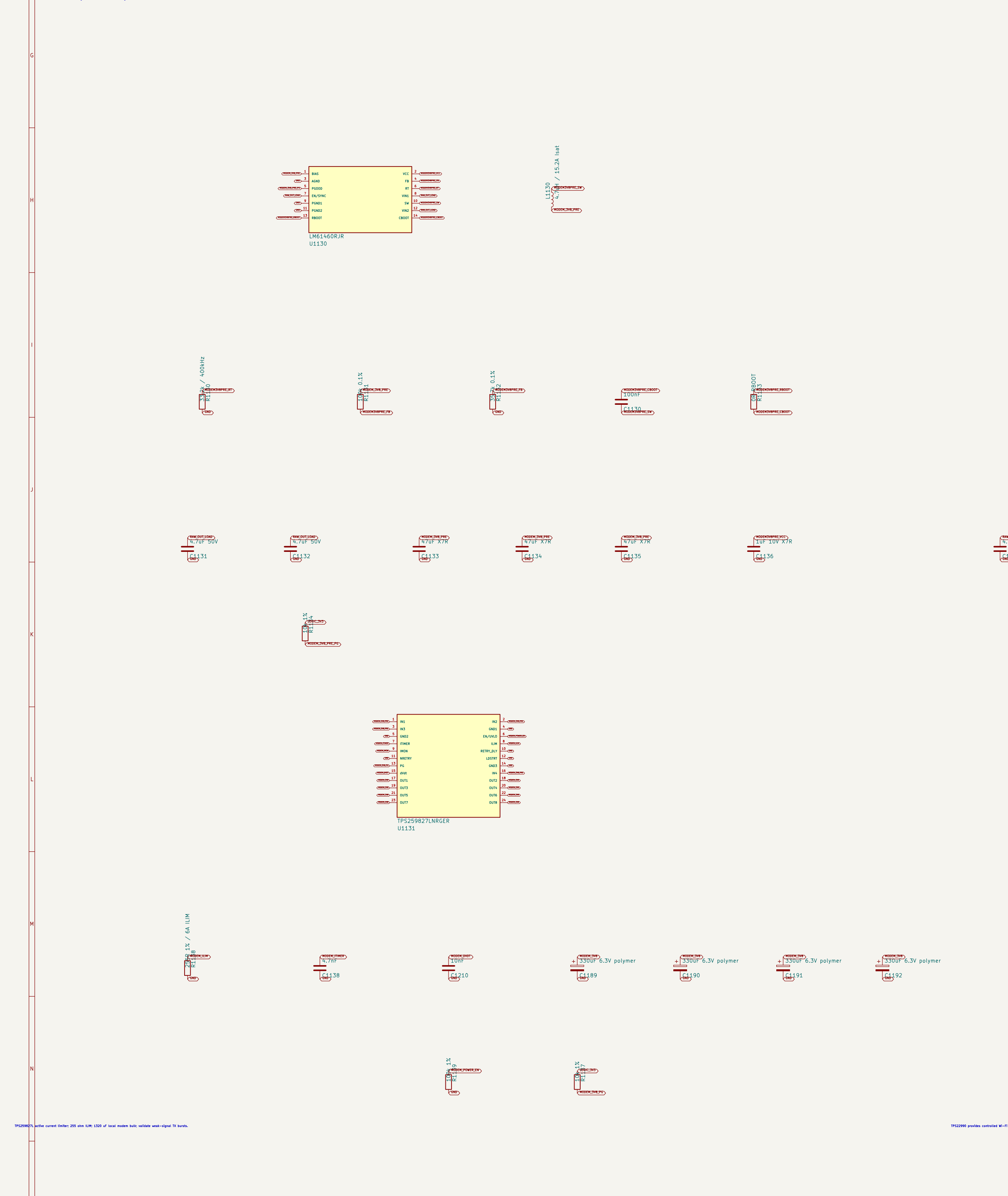
2. SYS_4V0 / 12 A SYNCHRONOUS BUCK



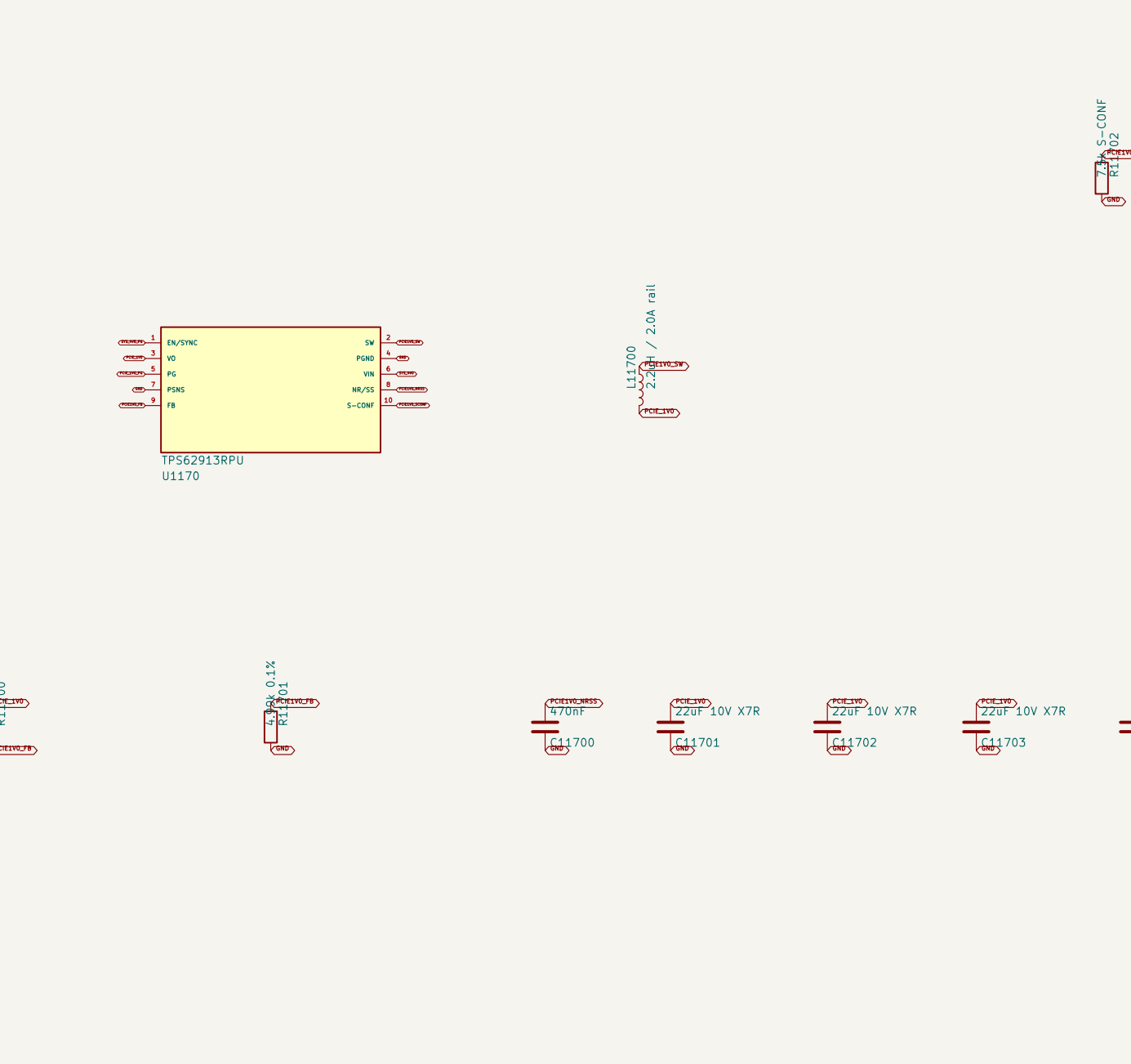
3. AUX_12V / 8 A FOUR-SWITCH BUCK-BOOST



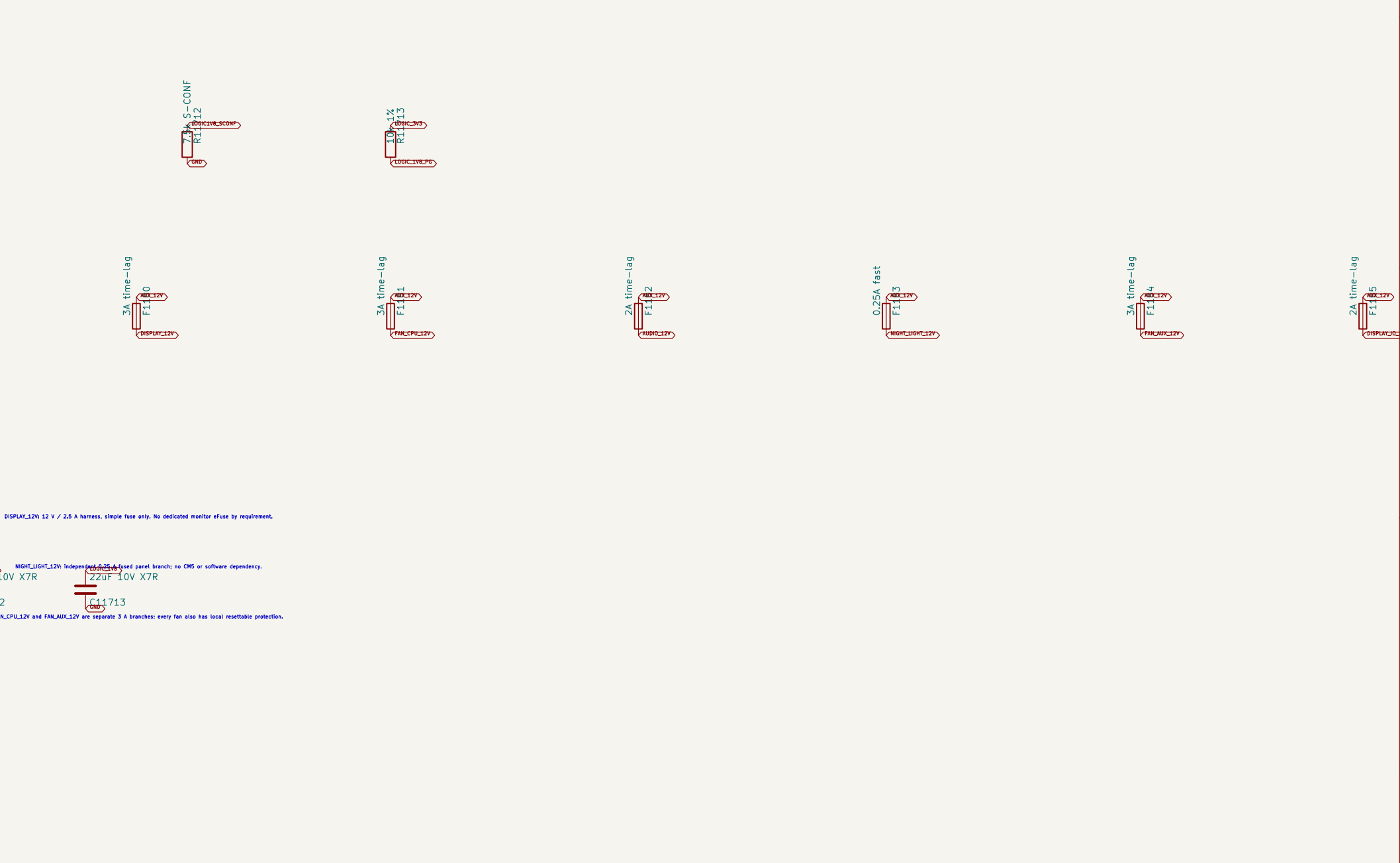
4. DEDICATED RADIO, NETWORK, AND LOGIC BUCKS



5. LOW-NOISE POINT-OF-LOAD RAILS



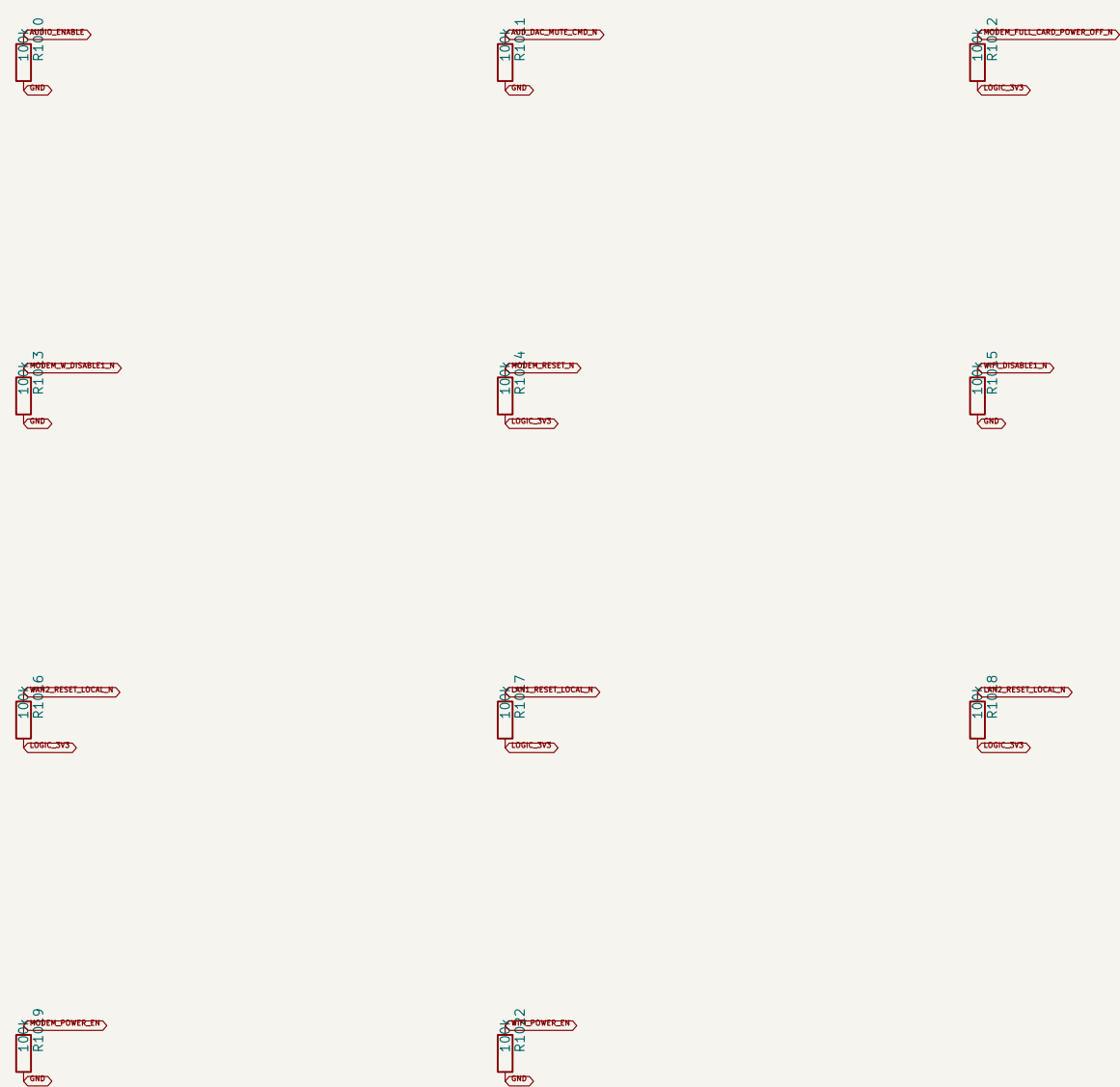
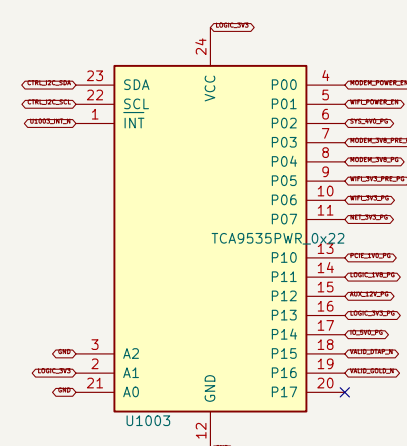
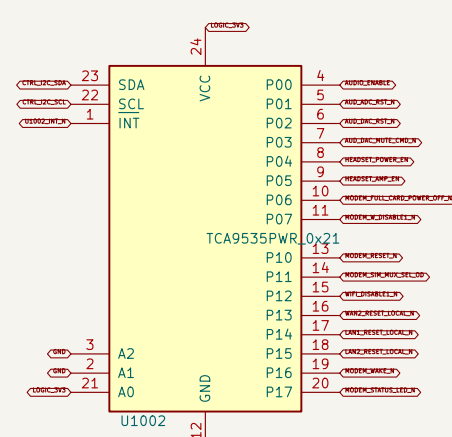
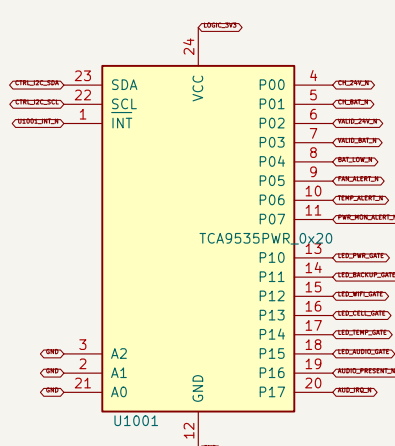
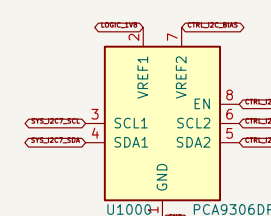
6. AUX_12V BRANCHES AND AUDIO HANDOFF



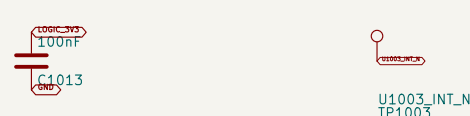
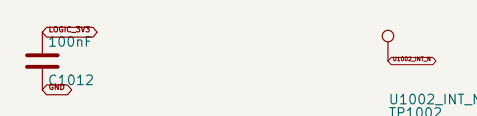
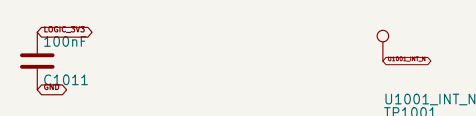
7. SEQUENCING, POWER=GOOD, AND TEST ACCESS

1. 1.8 V TO 3.3 V SYSTEM I2C CONTROL BUS

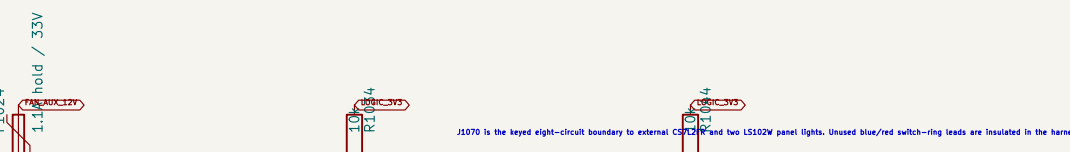
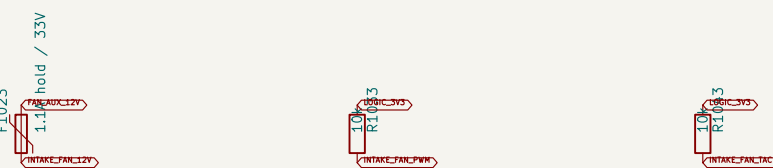
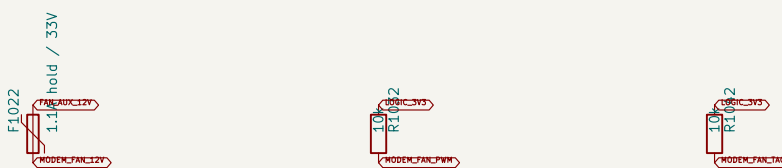
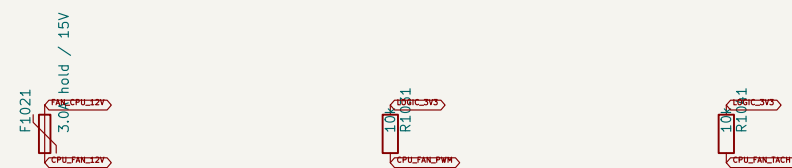
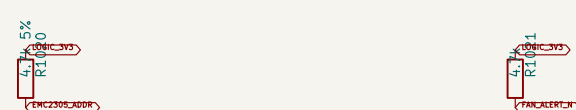
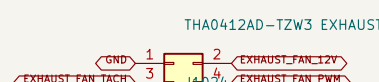
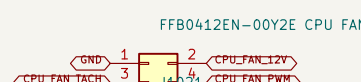
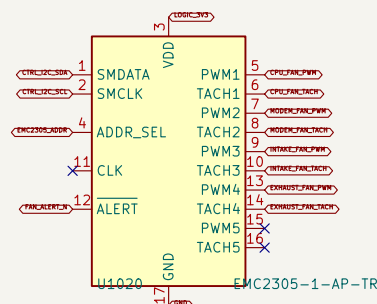
2. THREE 16-BIT GPIO EXPANDERS



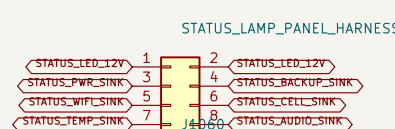
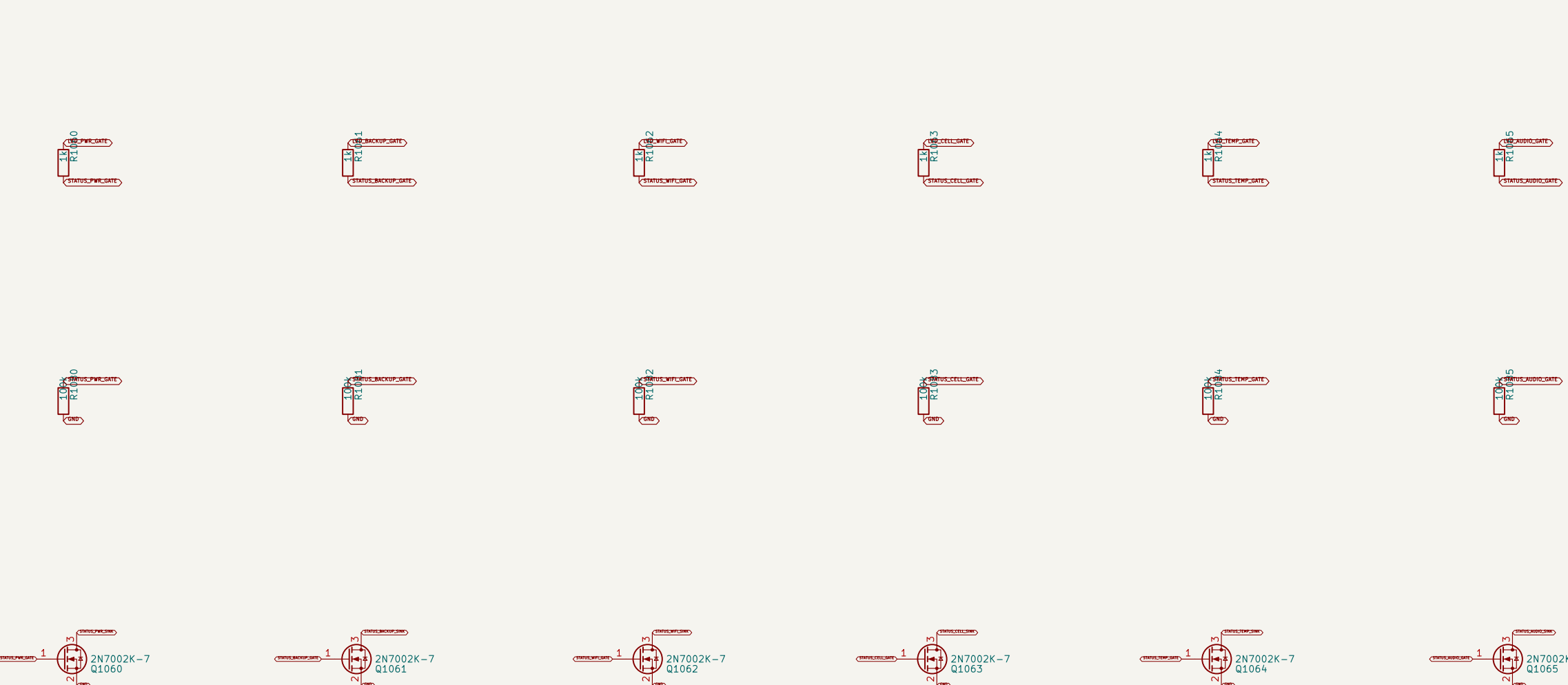
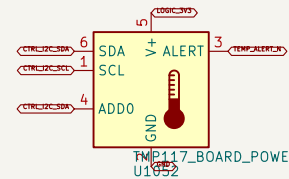
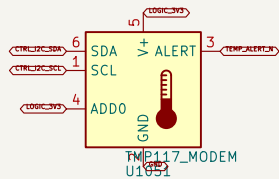
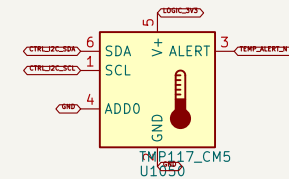
The control branch is isolated to its own PDB550, which is not in contact with the separate 4000-800 branch prepared



3. EMC2305 FOUR-FAN CONTROL WITH FAILSAFE



4. THREE HIGH-ACCURACY TEMPERATURE SENSORS



TC9A935 devices at 0x20/0x21/0x22 capture status and drive safe control output
TMP117 sensors monitor CMS, modem, and board/power zones at 0x48/0x49/0x4A
PWM pull-ups provide full-speed hardware fallback if the controller or firmware
EMC2305 controls CPU, modem, intake, and exhaust fans independently

ProComm Sheet: /Thermal / Fans / 10/ File: Thermal-10_kicad_sch		
Title: Radxa CM5 ProComm Carrier – Thermal and GPIO Cont		
Size: A0	Date: 2026-08-14	Rev: A1
KiCad E.D.A. 10.0.5-rc1		Id: 8/8